

0 9

Read the following passage.

The placenta is a specialised exchange surface.

In the placenta, substances are exchanged between the blood of a fetus and the blood of its mother. Gas exchange for the fetus occurs in the placenta.

There is also transfer of IgG antibodies in the placenta between the mother's blood and fetal blood. These IgG antibodies protect the fetus against the pathogens that infect its mother during pregnancy. The IgG antibodies can circulate at high concentration in the mother's blood for months or years. A fetus does not produce IgG antibodies. 5

The UK immunisation programme vaccinates as many babies as possible to protect the UK population against pathogens such as measles viruses and tetanus bacteria. Measles viruses spread quickly from infected people. Despite the efforts of the NHS, there has been a recent increase in the number of children catching measles. 10

Tetanus bacteria enter the body through skin wounds. Tetanus bacteria do not spread from infected people. In order to develop good immunity against tetanus, children are given three tetanus vaccinations at regular intervals before they reach their first birthday. 15

Use the information in the passage and your own knowledge to answer the following questions.

0 9 . 1

Gas exchange for the fetus occurs in the placenta (line 3).

Describe how the composition of blood in the pulmonary artery of a fetus is different from the composition of blood in the pulmonary artery of its mother.

Give **one** reason for this difference.

[2 marks]



0 9 . 2

Explain how a fetus is protected against the pathogens that infect its mother during pregnancy (lines 5–6).

Do **not** give details of an active immune response in the mother.

[3 marks]

0 9 . 3

Suggest how vaccinating as many babies as possible protects the UK **population** against pathogens such as measles viruses and tetanus bacteria (lines 9–11).

[2 marks]

Protection against measles _____

Protection against tetanus _____

Question 9 continues on the next page

Turn over ►



0 9 . 4

Suggest why there has been a recent increase in the number of children catching measles (lines 12–13).

[1 mark]

0 9 . 5

Explain why giving children more than one tetanus vaccination develops good immunity against tetanus (lines 15–17).

[2 marks]

10

END OF QUESTIONS

